

MGT 6078 Assignment 4

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View our code here (in addition to uploaded code file):
https://colab.research.google.com/drive/14_b-g_USwEM8sBwEdaLHbNaGkBMglgNp?usp=sharing

1 Replicate AQR Tables 1-4

Note: we present the t-statistic for each factor's beta value in the column next to said beta.

1.1 Table 1: FF5

	dependent	alpha	alpha_t	r2	beta_SMB	beta_SMB_t	beta_HML	beta_HML_t	beta_RMW	beta_RMW_t	beta_CMA	beta_CMA_t	beta_Mkt-RF	beta_Mkt-RF_t
0	Mkt-RF	0.785343	5.244589	0.200037	0.295003	5.637358	0.090297	1.314505	-0.254342	-3.554818	-0.818171	-8.268506	NaN	NaN
1	SMB	0.213363	2.042419	0.180348	NaN	NaN	0.168529	3.598099	-0.440699	-9.405759	-0.183783	-2.598811	0.139222	5.637358
2	HML	-0.047124	-0.579036	0.484611	0.101755	3.598099	NaN	NaN	0.176467	4.647774	1.015279	24.935397	0.025730	1.314505
3	RMW	0.365145	4.778189	0.158846	-0.241726	-9.405759	0.160310	4.647774	NaN	NaN	-0.241073	-4.648301	-0.065838	-3.554818
4	CMA	0.220234	4.114224	0.526508	-0.049080	-2.598811	0.449061	24.935397	-0.117374	-4.648301	NaN	NaN	-0.103116	-8.268506

1.2 Table 2: FF5 + Momentum

	dependent	alpha	alpha_t	r2	beta_SMB	beta_SMB_t	beta_HML	beta_HML_t	beta_RMW	beta_RMW_t	beta_CMA	beta_CMA_t	beta_Momentum	beta_Momentum_t	beta_Mkt-RF	beta_Mkt-RF_t
0	Mkt-RF	0.879878	5.892790	0.221370	0.289136	5.594873	0.008791	0.125250	-0.222541	-3.134936	-0.744180	-7.512539	-0.162473	-4.505734	NaN	NaN
1	SMB	0.209149	1.971289	0.180408	NaN	NaN	0.171393	3.536453	-0.441577	-9.388008	-0.185666	-2.606693	0.005908	0.232153	0.140182	5.594873
2	HML	0.049736	0.622170	0.517735	0.096840	3.536453	NaN	NaN	0.185453	5.043063	0.992418	25.097306	-0.132016	-7.134099	0.002408	0.125250
3	RMW	0.334246	4.308671	0.164065	-0.240722	-9.388008	0.178929	5.043063	NaN	NaN	-0.252514	-4.855188	0.040287	2.150704	-0.058818	-3.134936
4	CMA	0.188488	3.479336	0.532684	-0.048940	-2.606693	0.462979	25.097306	-0.122098	-4.855188	NaN	NaN	0.040622	3.129479	-0.095104	-7.512539
5	Momentum	0.710755	4.697201	0.101057	0.012310	0.232153	-0.486838	-7.134099	0.153985	2.150704	0.321113	3.129479	NaN	NaN	-0.164132	-4.505734

1.3 Table 3: Replace HML with HML-DEV in FF5

	dependent	alpha	alpha_t	r2	beta_SMB	beta_SMB_t	beta_HML_DEV	beta_HML_DEV_t	beta_RMW	beta_RMW_t	beta_CMA	beta_CMA_t	beta_Mkt-RF	beta_Mkt-RF_t
0	Mkt-RF	0.780924	5.250348	0.210639	0.282561	5.449208	16.730200	3.422963	-0.244078	-3.480269	-0.881936	-10.500029	NaN	NaN
1	SMB	0.212558	2.026914	0.174054	NaN	NaN	9.128174	2.682126	-0.418933	-8.987799	-0.102840	-1.648655	0.136179	5.449208
2	HML_DEV	-0.000609	-0.539555	0.299867	0.001052	2.682126	NaN	NaN	0.000528	1.002429	0.009867	17.473502	0.000929	3.422963
3	RMW	0.369064	4.763598	0.135529	-0.234357	-8.987799	2.562279	1.002429	NaN	NaN	-0.105766	-2.270670	-0.065805	-3.480269
4	CMA	0.277215	4.549671	0.383441	-0.035490	-1.648655	29.546929	17.473502	-0.065246	-2.270670	NaN	NaN	-0.146682	-10.500029

1.4 Table 4: FF5 with HML-DEV + Momentum

	dependent	alpha	alpha_t	r2	beta_SMB	beta_SMB_t	beta_HML_DEV	beta_HML_DEV_t	beta_RMW	beta_RMW_t	beta_CMA	beta_CMA_t	beta_Momentum	beta_Momentum_t	beta_Mkt-RF	beta_Mkt-RF_t
0	Mkt-RF	0.881059	5.830447	0.221354	0.290183	5.624743	-0.221792	-0.030823	-0.220718	-3.149325	-0.733486	-7.676524	-0.164811	-3.193342	NaN	NaN
1	SMB	0.166108	1.543841	0.177938	NaN	NaN	15.949671	3.200519	-0.425191	-9.113907	-0.158372	-2.295776	0.067644	1.871107	0.141110	5.624743
2	HML_DEV	0.003274	4.198847	0.675889	0.000855	3.200519	NaN	NaN	0.000888	2.475836	0.008950	23.203969	-0.005292	-29.320323	-0.000006	-0.030823
3	RMW	0.321764	4.042469	0.142468	-0.237064	-9.113907	9.238168	2.475836	NaN	NaN	-0.159561	-3.106774	0.065988	2.448644	-0.059842	-3.149325
4	CMA	0.072687	1.272446	0.499282	-0.044595	-2.295776	47.019792	23.203969	-0.080585	-3.106774	NaN	NaN	0.226886	13.093084	-0.100435	-7.676524
5	Momentum	0.671900	6.313282	0.555272	0.069519	1.871107	-101.479349	-29.320323	0.121637	2.448644	0.828091	13.093084	NaN	NaN	-0.082367	-3.193342

2 Analysis

2.1 25 Portfolios Formed on Size and Book-to-Market

(a) Fama and French Five Factors (FF5 Model)

	portfolio	alpha_monthly	alpha_annualized	alpha_tstat
0	SMALL LoBM	-0.265235	-3.182825	-3.104871
1	ME1 BM2	0.151294	1.815527	2.283398
2	ME1 BM3	0.017480	0.209758	0.328902
3	ME1 BM4	0.152416	1.828991	3.105963
4	SMALL HiBM	0.150937	1.811243	2.051324
5	ME2 BM1	-0.078155	-0.937862	-1.312461
6	ME2 BM2	0.018577	0.222927	0.374552
7	ME2 BM3	0.008415	0.100984	0.164182
8	ME2 BM4	0.023616	0.283394	0.505823
9	ME2 BM5	-0.022401	-0.268815	-0.437229
10	ME3 BM1	-0.018836	-0.226036	-0.334908
11	ME3 BM2	0.004039	0.048468	0.073359
12	ME3 BM3	-0.084636	-1.015626	-1.510084
13	ME3 BM4	-0.018098	-0.217180	-0.335383
14	ME3 BM5	-0.029290	-0.351482	-0.434367
15	ME4 BM1	0.124475	1.493694	2.140870
16	ME4 BM2	-0.145181	-1.742168	-2.413697
17	ME4 BM3	-0.123915	-1.486979	-2.008968
18	ME4 BM4	0.035093	0.421112	0.540156
19	ME4 BM5	-0.099028	-1.188330	-1.283276
20	BIG LoBM	0.115029	1.380352	2.966050
21	ME5 BM2	-0.088678	-1.064135	-1.644955
22	ME5 BM3	-0.087513	-1.050157	-1.365245
23	ME5 BM4	-0.218999	-2.627983	-3.670988
24	BIG HiBM	-0.023561	-0.282729	-0.258234

GRS Statistic: 2.921503
p-value: 0.000003

Interpretation: With a very low p-value (0.000003), which is significantly less than common significance levels (e.g., 0.05), we reject the null hypothesis that the alphas for the 25 portfolios are jointly equal to zero. This suggests that the Fama and French Five Factors do not fully explain the returns on these test assets, indicating the presence of significant mispricing or uncaptured risk factors by the FF5 model for these portfolios.

(b) AQR Six-Factor Model (FF5 with HML-DEV + Momentum)

	portfolio	alpha_monthly	alpha_annualized	alpha_tstat
0	SMALL LoBM	-0.163074	-1.956886	-1.793542
1	ME1 BM2	0.166397	1.996767	2.427663
2	ME1 BM3	0.014839	0.178063	0.271456
3	ME1 BM4	0.085146	1.021756	1.594016
4	SMALL HiBM	0.072426	0.869107	0.926482
5	ME2 BM1	0.037488	0.449858	0.574197
6	ME2 BM2	0.068441	0.821295	1.356144
7	ME2 BM3	-0.002896	-0.034751	-0.053675
8	ME2 BM4	-0.051428	-0.617132	-0.960552
9	ME2 BM5	-0.165571	-1.986856	-2.756480
10	ME3 BM1	0.083150	0.997799	1.370461
11	ME3 BM2	0.011811	0.141736	0.209370
12	ME3 BM3	-0.104737	-1.256841	-1.752887
13	ME3 BM4	-0.100985	-1.211826	-1.667540
14	ME3 BM5	-0.151351	-1.816216	-1.926880
15	ME4 BM1	0.206808	2.481697	3.384557
16	ME4 BM2	-0.122558	-1.470698	-1.998025
17	ME4 BM3	-0.184087	-2.209046	-2.893458
18	ME4 BM4	-0.060061	-0.720732	-0.869376
19	ME4 BM5	-0.206800	-2.481595	-2.394812
20	BIG LoBM	0.205350	2.464199	5.027401
21	ME5 BM2	-0.102711	-1.232530	-1.860952
22	ME5 BM3	-0.178709	-2.144510	-2.796928
23	ME5 BM4	-0.361606	-4.339270	-5.383405
24	BIG HiBM	-0.171758	-2.061092	-1.571861

GRS Statistic: 4.056111
p-value: 2.577620e-10

Interpretation: Similar to the FF5 model, the AQR six-factor model also yields a very low p-value (2.577620×10^{-10}). Therefore, we reject the null hypothesis that the alphas for the 25 portfolios are jointly equal to zero. This implies that even with the inclusion of the HML-DEV and Momentum factors, this six-factor model does not fully explain the returns on these test assets, suggesting remaining unexplained variations or mispricing.

2.2 10 Industry Portfolios

(a) Fama and French Five Factors (FF5 Model)

	portfolio	alpha_monthly	alpha_annualized	alpha_tstat
0	NoDur	-0.119539	-1.434467	-1.495722
1	Durbl	-0.179834	-2.158006	-0.981304
2	Manuf	-0.212065	-2.544774	-3.566215
3	Enrgy	-0.163835	-1.966022	-0.978019
4	HiTec	0.397707	4.772490	4.024413
5	Telcm	-0.013933	-0.167199	-0.120635
6	Shops	-0.008966	-0.107598	-0.098866
7	Hlth	0.140069	1.680828	1.265017
8	Utils	-0.048703	-0.584435	-0.417247
9	Other	-0.195956	-2.351469	-3.214300

GRS Statistic: 4.083256
p-value: 0.000017

Interpretation: We would again reject the null hypothesis, based on the reasons given above (very low p-value of 0.000017).

(b) AQR Six-Factor Model (FF5 with HML-DEV + Momentum)

	portfolio	alpha_monthly	alpha_annualized	alpha_tstat
0	NoDur	-0.087520	-1.050237	-1.068400
1	Durbl	-0.075147	-0.901763	-0.415086
2	Manuf	-0.196488	-2.357850	-3.217273
3	Enrgy	-0.373116	-4.477393	-2.213935
4	HiTec	0.542073	6.504880	5.337613
5	Telcm	-0.023813	-0.285755	-0.205083
6	Shops	0.088383	1.060596	0.960029
7	Hlth	0.208502	2.502019	1.853562
8	Utils	-0.179642	-2.155700	-1.522116
9	Other	-0.252578	-3.030935	-3.738412

GRS Statistic: 5.897026

p-value: 1.321323e-08

Interpretation: With a very low p-value (1.321323e-08), we reject the null hypothesis that the alphas for the 10 industry portfolios are jointly equal to zero. This indicates that the AQR six-factor model does not fully explain the returns on these industry portfolios, suggesting the presence of significant uncaptured factors or mispricing within these industry groupings.

2.3 Final Thoughts

We note that the GRS test indicates that we reject the null hypothesis that the alphas are jointly equal to zero for all of these factor models. Once adding HML-DEV and Momentum to the FF5 factor model for both types of portfolio classes, the p-value becomes smaller, not larger, and the GRS statistic rises. This implies that the added factors actually make the model fit look worse under the GRS test lens. Model complexity doesn't guarantee better joint pricing performance.